BIODIVERSITY AND CONSERVATION

WHAT IS BIODIVERSITY

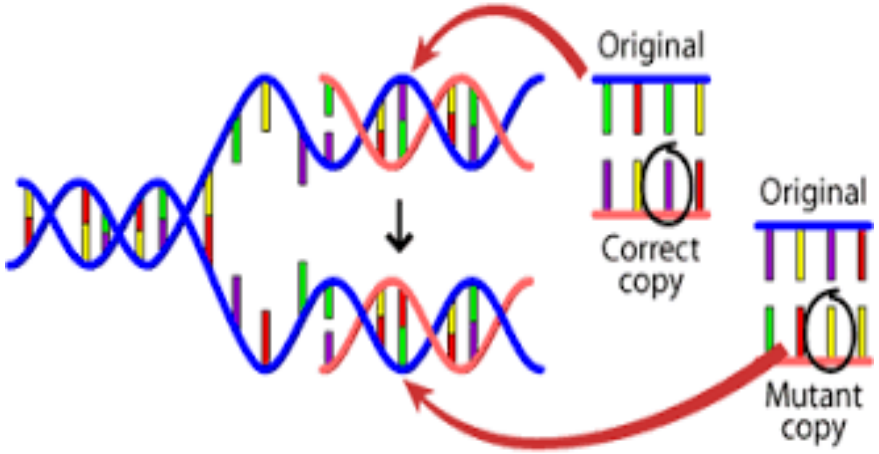
- The word “biodiversity” is a contraction of “biological diversity”.
- The father of biodiversity Edward O. Wilson (an eminent entomologist) first coined this term in 1986.
- Diversity is a vast concept refers to the range of variations or differences among some set of entities; biological diversity thus refers to varieties within the living world.
- The term 'biodiversity' is generally considered as an ‘Umbrella term’ referring to organisms found within the living world.
- It is commonly used to describe the number, variety of life and variability of living organisms.

ELEMENTS OF BIO-DIVERSITY

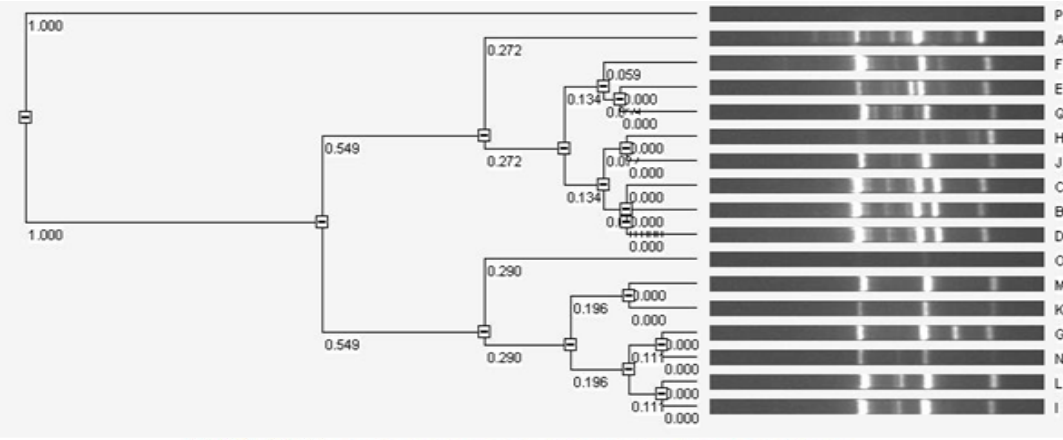
- The predicted number of total species varies from 5 to 50 million and averages at 14 million.
- Out of the total known species, about 60 % are insects, about 16 % are higher plants and only about 0.3 % is mammals.
- The most unique feature of Earth is the existence of life and the most extraordinary feature of life is its diversity.
- Bio-diversity is normally treated in terms of genes, species and ecosystems in correspondence with three fundamental hierarchical levels of biological organization.
- The three diversities are referred as genetic, species and ecosystem diversity.

A. GENETIC DIVERSITY

- Referred as the diversity within the species or intra or infra species diversity.
- It encompasses the components of genetic coding that structures organisms (nucleotides, genes, chromosomes) and variation in the genetic make-up between individuals within a population and between populations.
- The variation could be in alleles (different variants of same genes), in entire genes (the traits determining particular characteristics) or in chromosomal structures.
- This enables a population to adapt to its environment and to respond to natural selection.



- If a species has more genetic diversity, it can adapt better to the changed environmental conditions.
- Lower diversity in a species leads to uniformity and the amount of genetic variation is the basis of speciation (evolution of new species).
- It has a key role in the maintenance of diversity at species levels.
- Genetic diversity within a species often increases with environmental variability.
- It can be measured by the methods based on DNA marker like RFLP, PCR, RAPD, APPCR, PFGE, microsatellite primed PCR and others.



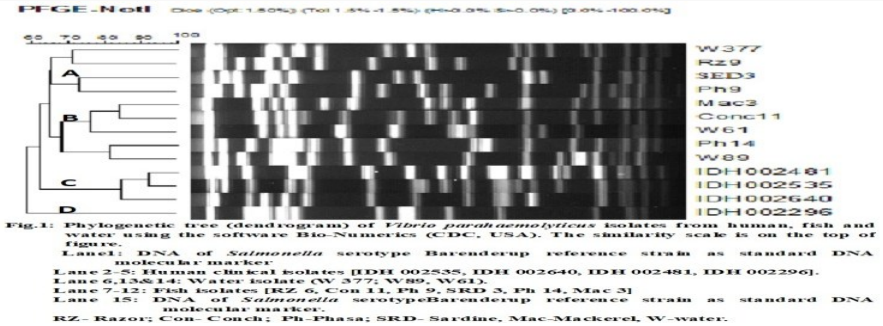
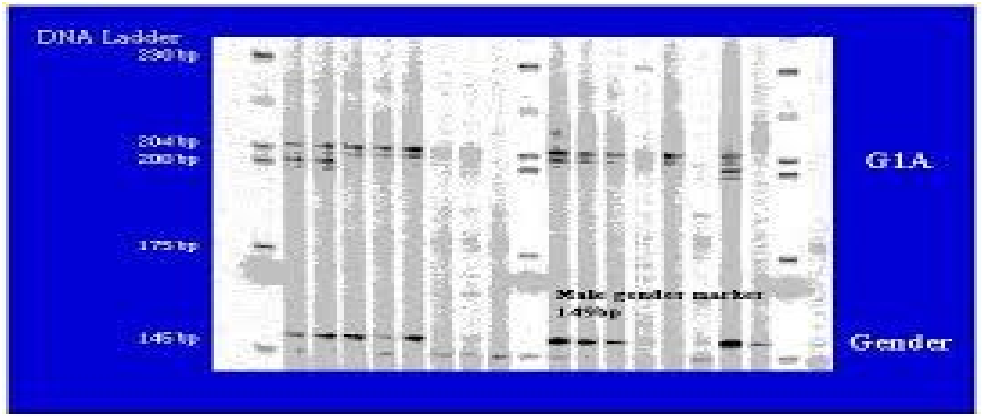
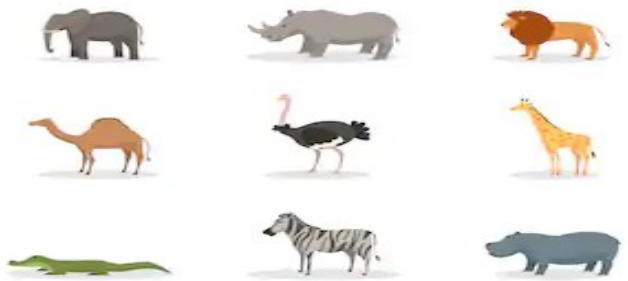


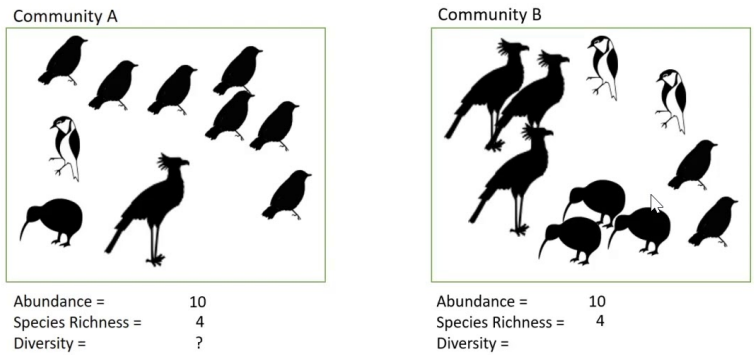
Fig.1: Phylogenetic tree (dendrogram) of *Vibrio parahaemolyticus* isolates from human, fish and water using the software Bio-Numerics (CDC, USA). The similarity scale is on the top of figure.
Lane1: DNA of *Salmonella* serotype Barenderup reference strain as standard DNA molecular marker.
Lane 2-5: Human clinical isolates [IDH 002535, IDH 002640, IDH 002481, IDH 002296].
Lane 6,13&14: Water isolate (W 377; W89, W61).
Lane 7-12: Fish isolates [RZ 6, Con 11, Ph 9, SKD 3, Ph 14, Mac 3]
Lane 15: DNA of *Salmonella* serotypeBarenderup reference strain as standard DNA molecular marker.
RZ- Razor; Con- Conch; Ph-Phasa; SKD- Sardine, Mac-Mackerel, W-water.

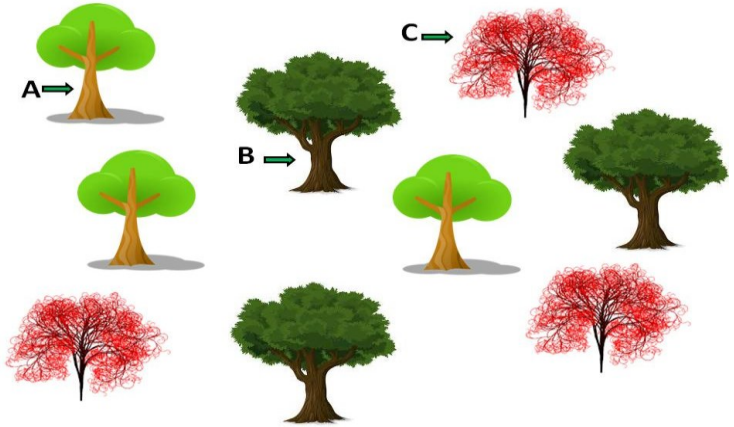


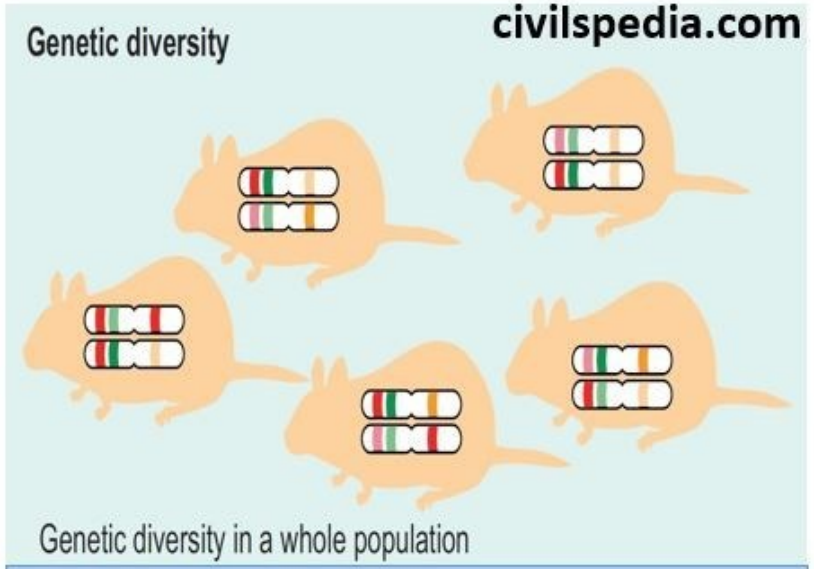
B. SPECIES DIVERSITY

- Species are distinct units of diversity, each playing a specific role in an ecosystem.
- Refers to the variety of living species within a geographic area.
- It encompasses the taxonomic hierarchy and its components, from individuals upwards to species, genera and beyond.
- The simplest measures of species diversity (SR+SE) are:
  - Species richness (the number of species per unit area)
  - Species evenness (the evenness in number of individuals of each species in the area).
- Generally, greater the species richness greater is the species diversity.









Genetic Diversity

Species Diversity

Ecosystem Diversity

C. ECOSYSTEM DIVERSITY

Ecosystem diversity addresses the combined characteristics of biotic properties (biodiversity) and abiotic properties (geodiversity).

It is a variation in the ecosystems found in a region or the variation in ecosystems over the whole planet.

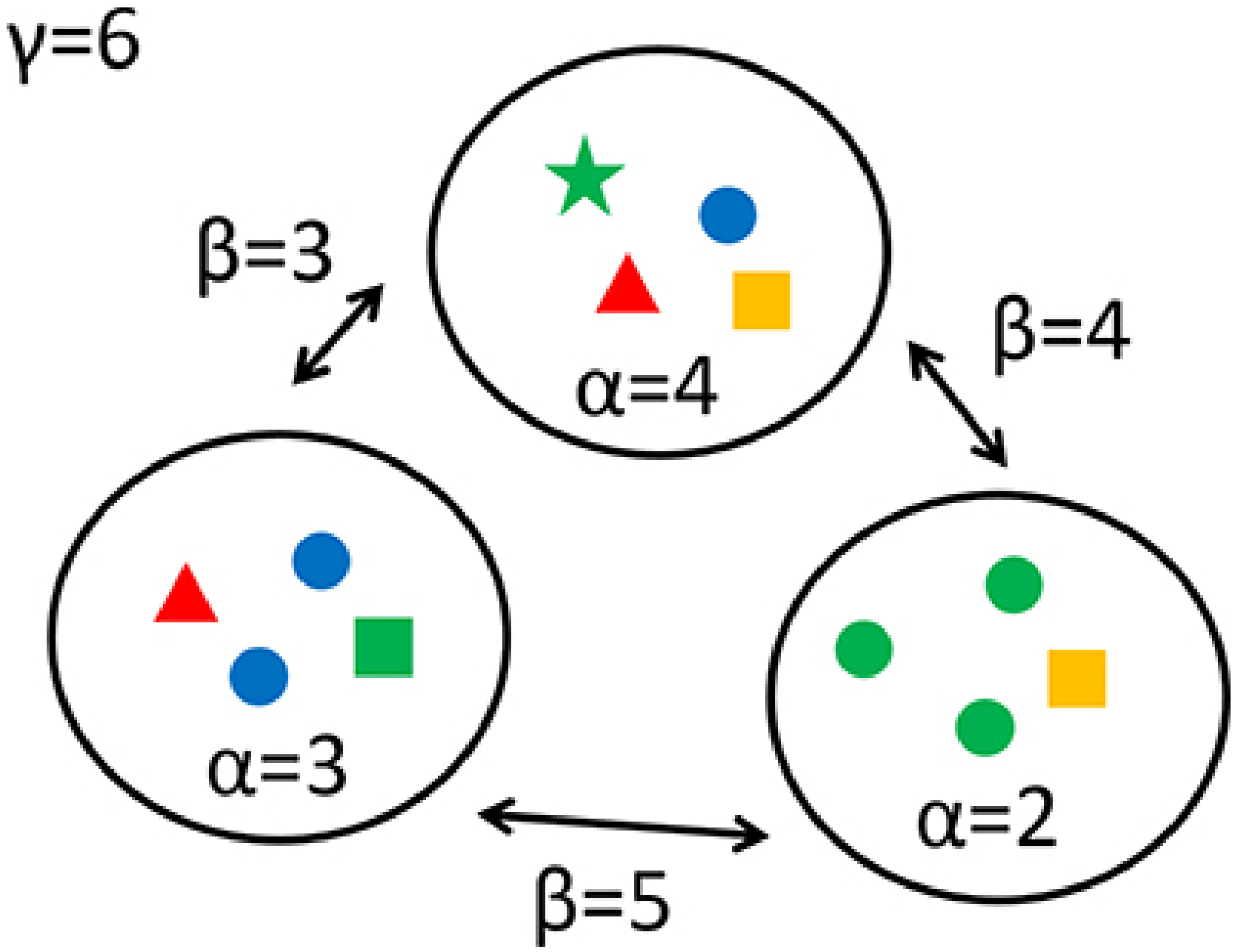
An example of ecological diversity on a global scale would be the variation in ecosystems, such as deserts, forests, grasslands, wetlands and oceans.

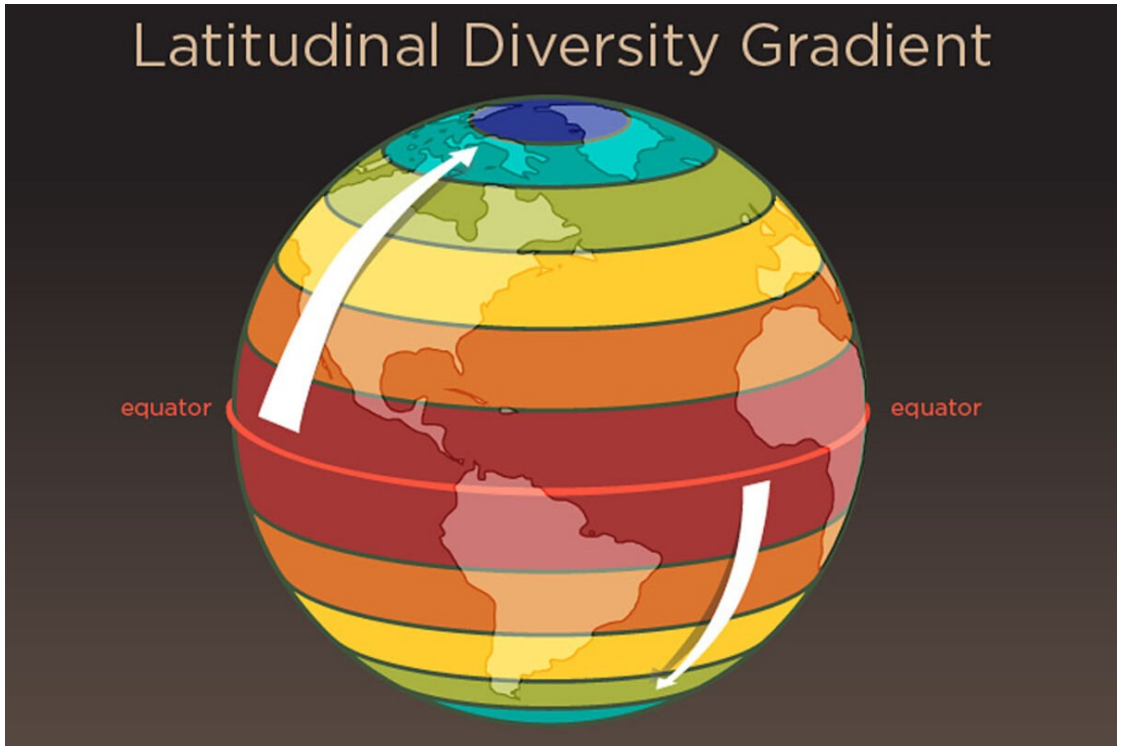
ALPHA, BETA, AND GAMMA DIVERSITY

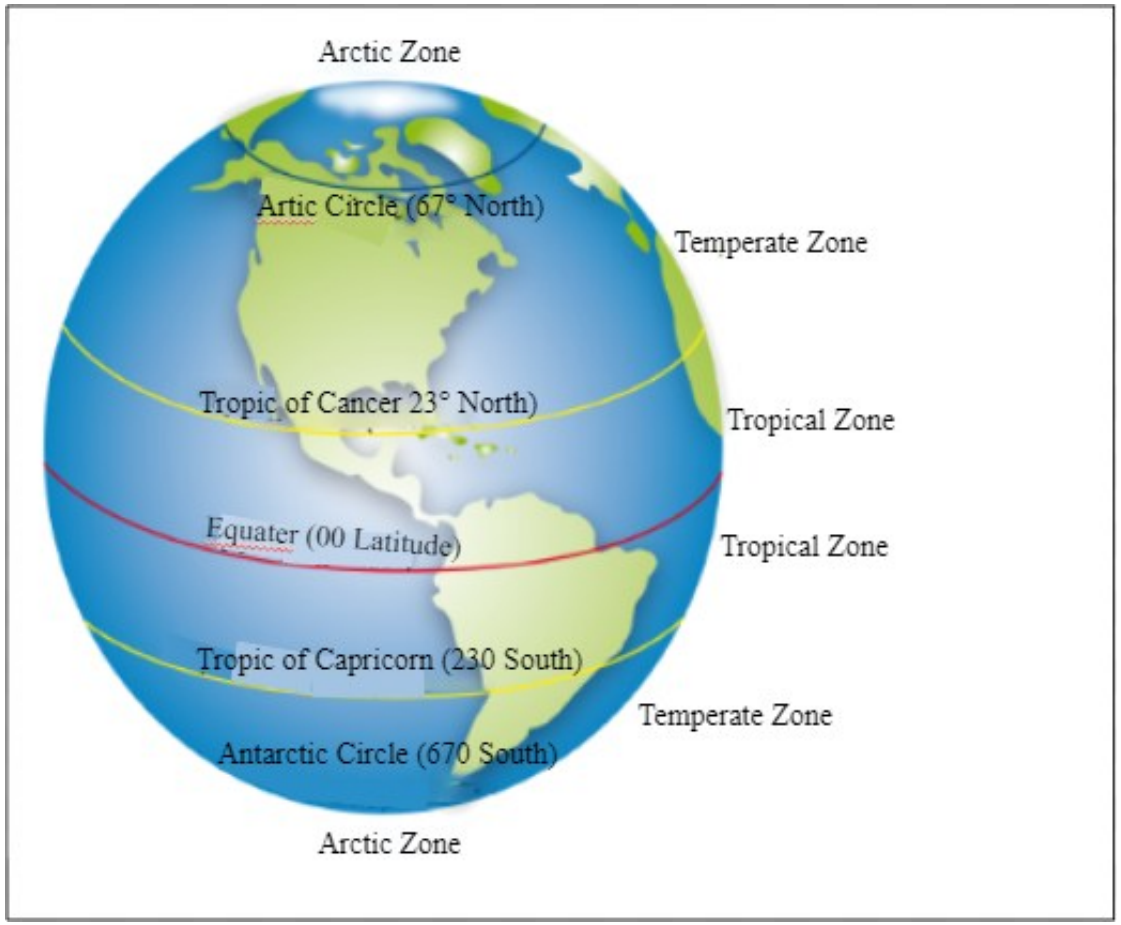
- Alpha Diversity-
  Refers to the diversity of organisms sharing the same community or habitat. A combination of species richness and evenness is used to represent diversity within a community or habitat

- Beta Diversity-
  Indicates diversity between communities. Species frequently change when habitat or community changes.

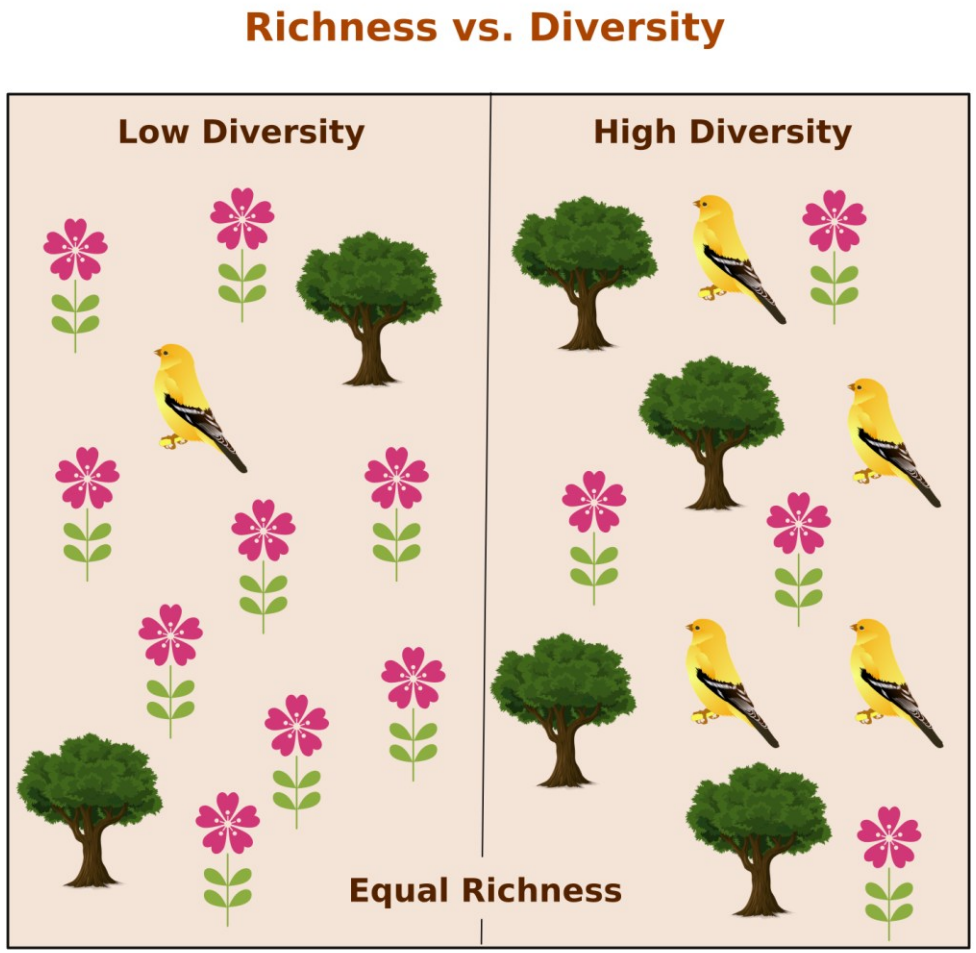
- Gamma Diversity-
  Refers to the diversity of the habitats over the total landscape or geographical area.


γ=6
β=3
β=4
α=4
α=3
α=2
β=5


Latitudinal Diversity Gradient
equator
equator


Arctic Zone
Artic Circle (67° North)
Temperate Zone
Tropic of Cancer 23° North)
Tropical Zone
Equater (00 Latitude)
Tropical Zone
Tropic of Capricorn (230 South)
Temperate Zone
Antarctic Circle (670 South)
Arctic Zone

Species richness is a measure of the number of different species in a given area or ecosystem. It is a simple count and does not consider the relative abundance of each species.



SPECIES–AREA RELATIONSHIP

- The SAR is derived from island biogeography theory. It is **an empirically measured relationship between the size of an area and the number of species it contains**. The larger the area, the greater the number of species present.
- The species–area relationship or species–area curve describes the relationship between the area of a habitat, or of part of a habitat, and the number of species found within that area.
- The species–area relationship is usually constructed for a single type of organism, such as all vascular plants or all species of a specific trophic level within a particular site.
- As the area of a series of islands increase, there is a direct relationship to the increasing species richness of primary producers

The species-area relationship equation is: $S = cA^z$

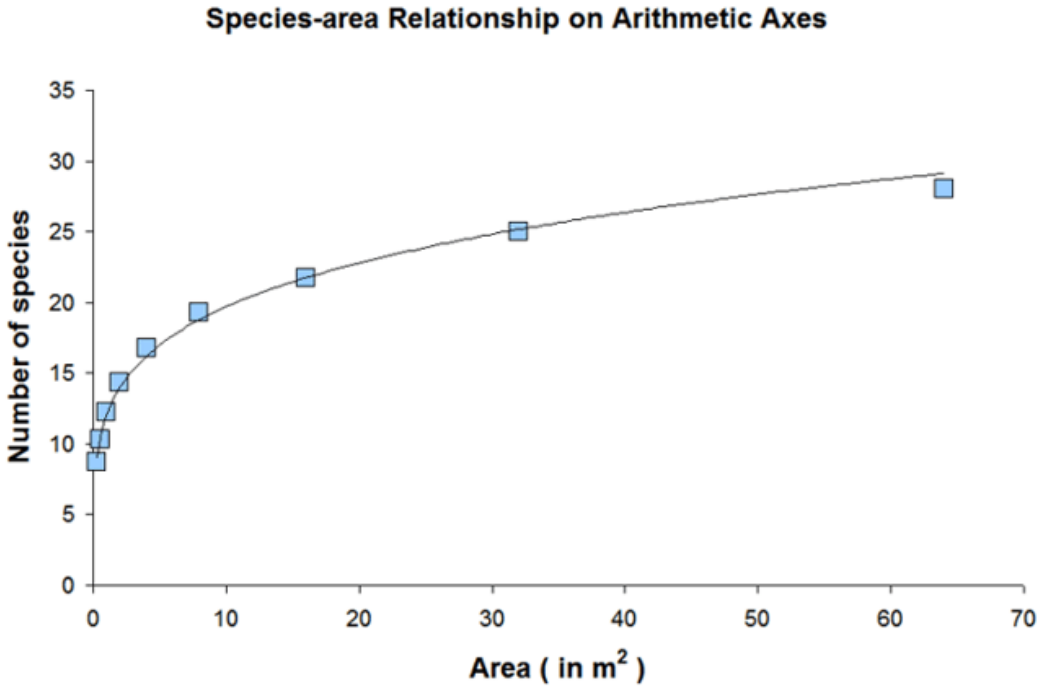
In this equation,

S represents the measure of diversity of a species (for example, the number of species)
c is a constant representing the y-intercept.
A represents the area of the island or space that is being examined
z represents the slope of the area curve.

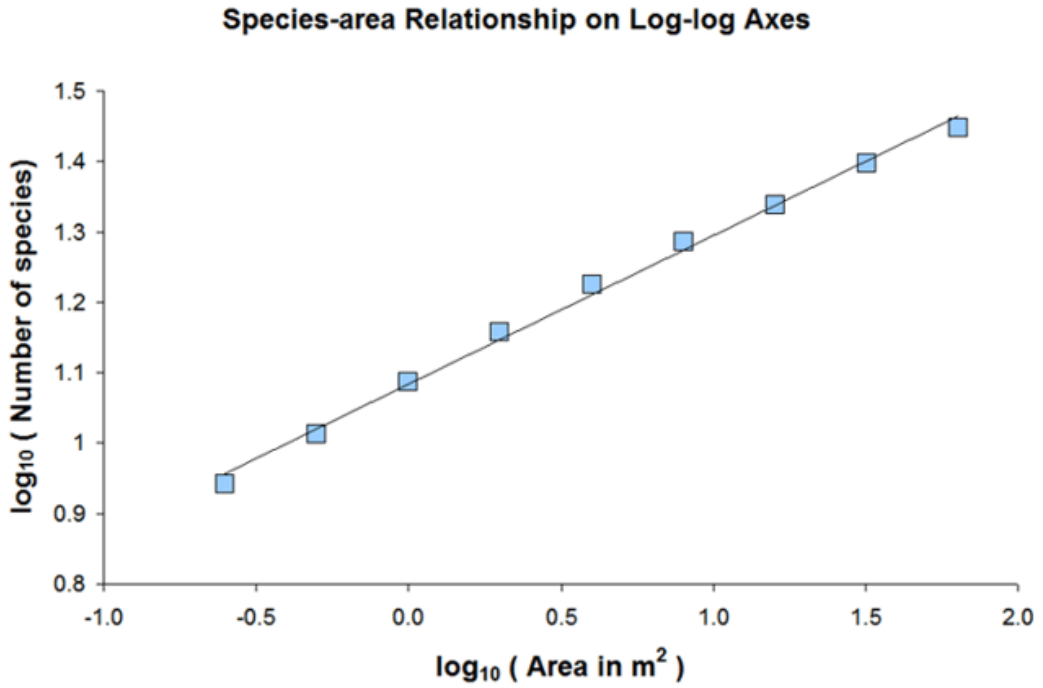
This function can also be expressed as a logarithmic function: log (S) = log(c) + z log (A)

This logarithmic expression of the function allows for the function to be drawn as a linear function. However, the core meaning of the function is the same: the area of the island dictates the species area relationship.

Alexander Van Humboldt gave the concept of Specie-Area relationship based upon his observation in South American jungle studies. He observed that within a region the richness of specie increases with the increase in explored area, but only up to a limit.



Ecologist describes the value of Z is in range of 0.1 and 0.2 irrespective of the specie or region evaluated. But, if we find the species - area relationship for a very large area like that of a continent then the value of Z is observed to be between 0.6 and 1.2 thus the graph in such situation comes out to be much steeper.



MCQs.

1. The curve of species area relationship looks like -

a. Sigmoidal

b.Elliptical

c.Rectangular hyperbola

d.None of the above

Fact: The relationship between species richness and area for a number of taxa like angiospermic plants, fresh water fishes and birds is found to be a rectangular hyperbola.

Ans.c

2.In larger area like continents slope tends to be-

a. Similar

b.Steeper

c.Both a and b

d.None of the above

Ans.b

3.In a species-area relationship -

a. the number of species in a sample increases as the area covered by that sample increases.

b.the number of species in an area of habitat increases as the area of that habitat increases.

c. the type of species observed in a habitat alters as the area of habitat alters.

d. species diversity increases linearly with island size.

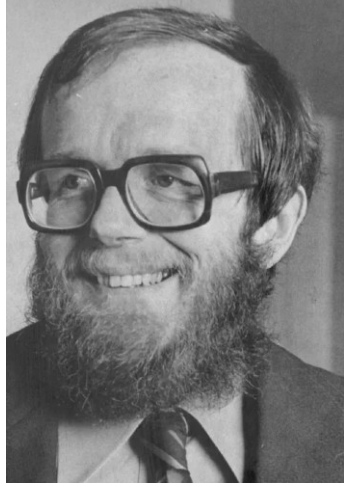
Ans. b

4.Species richness tends to decrease with increasing altitude. Which of the following is *not* believed to play some part in this trend?
a. Higher-altitude areas tend to be younger, in evolutionary terms.
b.Higher-altitude areas tend to occupy relatively small areas.
c.Productivity declines with altitude.
d.High-altitude habitats tend to be isolated from one another.

Ans.a

5.Which of the following plants have higher species diversity ?
a.Bryophytes
b.Algae
c.Fungi
d.Angiosperms
Ans. c

BIODIVERSITY HOTSPOTS



- A **biodiversity hotspot** is a biogeographic region with significant levels of biodiversity that is threatened by human habitation.
- Norman Myers wrote about the concept in two articles in *The Environmentalist* in 1988. *Hotspots: Earth's Biologically Richest and Most Endangered Terrestrial Ecoregions*
- Biodiversity hotspots host their diverse ecosystems on just 2.4% of the planet's surface.

Important Terms

- Vascular Plants: A plant that is characterized by the presence of conducting tissue (like xylem, Phloem, etc.)
- Endemic Species: A species unique to a particular bio-geographic location and not found elsewhere in the world. Ex. Habenaria panigrahiana, Lasiococca comberi, Sapria Himalayana, Ovaria Lurida are endemic to India.

Criteria

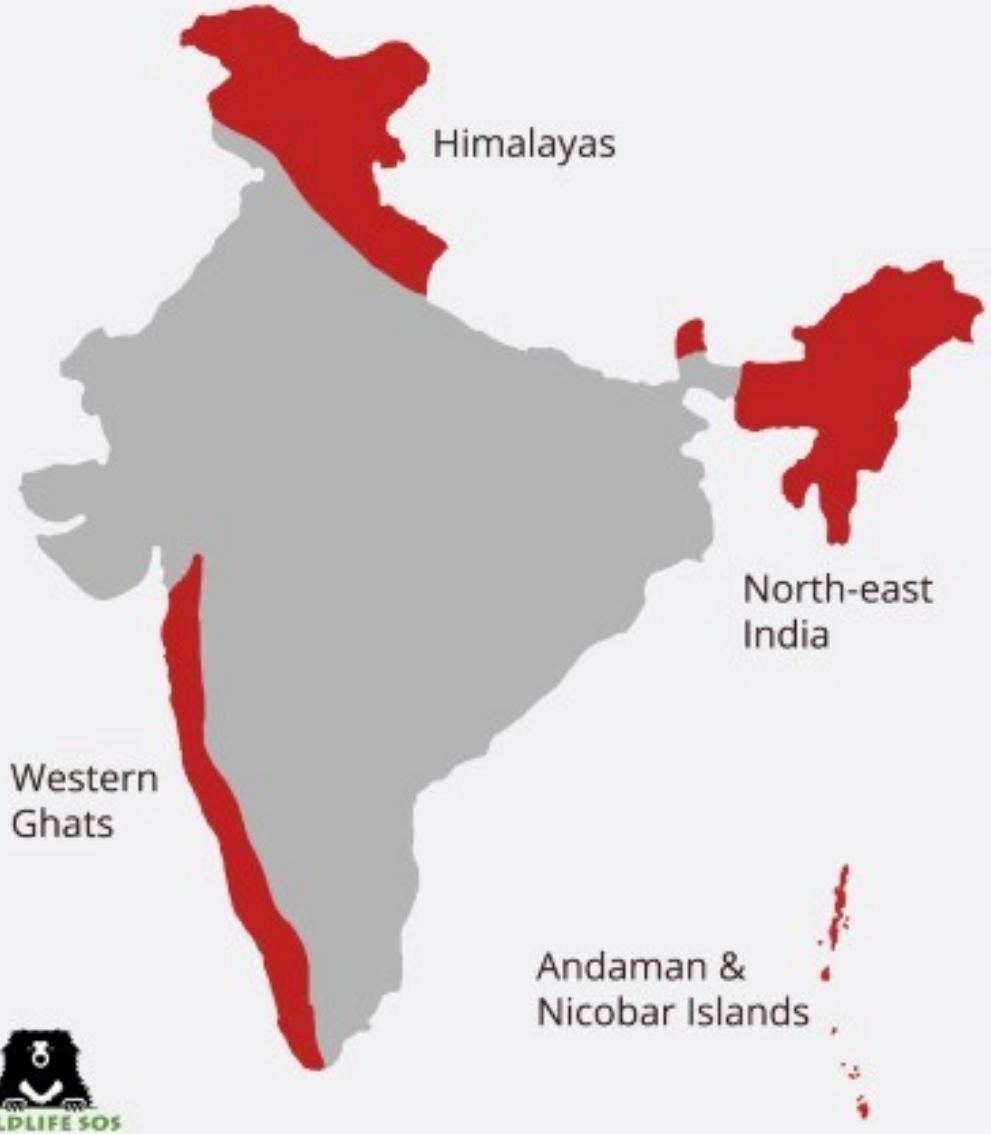
To qualify as a biodiversity hotspot, a region must meet two strict criteria:

1. It must have at least 1,500 vascular plants as endemics (i.e. 0.5% of the world's total of 3,00,000).

2. It must have ≤30% of its original natural vegetation left.

https://www.pbslearningmedia.org/resource/wim-sci-timeline/building-a-biodiversity-timeline/

INDIA'S BIODIVERSITY HOTSPOTS IN BRIEF



Total number of biodiversity hotspots: **4**

Total combined geographical area- **24.46%**

WILDLIFE SOS

USES OF BIODIVERSITY

- **The different types of the direct-use value of biodiversity are for food, medicine, biological control, industrial materials, recreational harvesting and ecotourism.**

- **Food: in form of vegetables, fruit, nuts, meat, milk as well as food colorants, flavoring and preservatives.**

- **Medicine: Willow trees (salicylic acid; aspirin), Foxglove (digitoxin), Atropa belladonna (atropine), Opium sativum (codeine), Papaver somniferum (morphine), Chinchona ledgeriana (quinine).** More than 25 per cent of the drugs currently sold in the market worldwide are derived from plants and 25,000 species of plants contribute to the traditional medicines used by native peoples around the world. Nobody knows how many more medicinally useful plants there are in tropical rain forests waiting to be explored

- **Animals source products (e.g. anticoagulants, coagulants, vasodilator agents) and for models on which to test potentially useful drugs or techniques.**

- **Biological control: Vector Control using biological aget**

- **Industrial materials: building materials, fibers, dyes, resins, gums, adhesives, rubber, oils and waxes, agricultural chemicals (including pesticides) and perfumes.**

- **Recreational harvesting: It is the harvesting of animals (e.g. fish, reptiles, birds, mammals) for display and as pets.**

THREATS TO BIODIVERSITY

The global change in the factors responsible for biodiversity results into loss of biodiversity.

Primary Factors:

- **Habitat loss and Fragmentation**
- **Greater climatic changes & global warming,**
- **Pollution,**
- **Over-exploitation of natural resources,**
- **Invasive species,**
- **Genetic manipulation of species.**

The secondary factors:

- **Population growth,**
- **Unsustainable patterns of consumption,**
- **Increased waste production,**
- **Urban development and**
- **International conflict.**

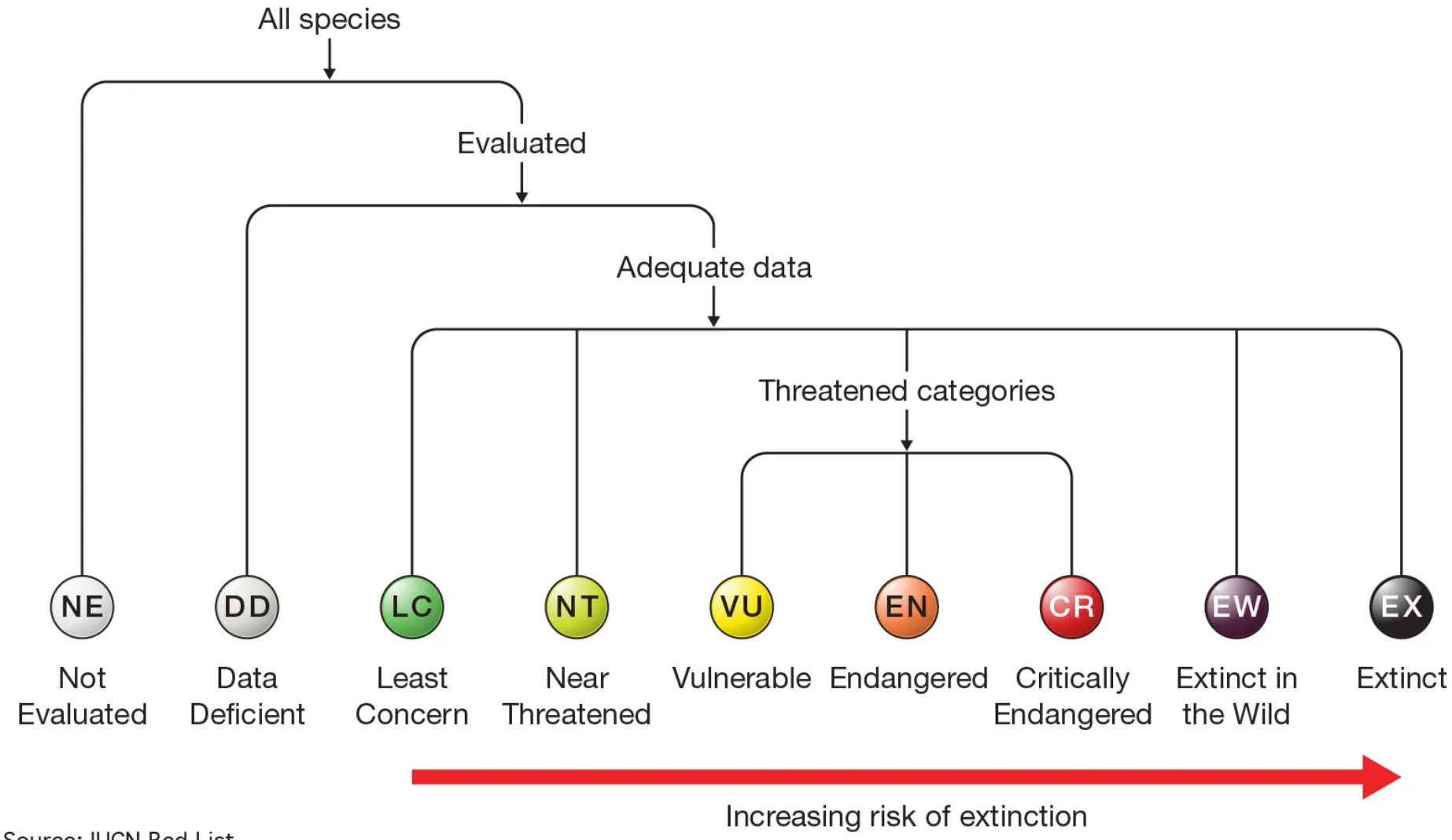
BASELINE DATA ON BIODIVERSITY

- **Red Data Book:** The **IUCN** published a Red Data Book in 2000 where categorization of various species into extinct, endangered, vulnerable and rare was done. It is updated frequently.

- **Green Book:** The Botanical Survey of India **(BSI)** published a Green Book which enlists rare plants growing in Protected areas like botanical gardens.

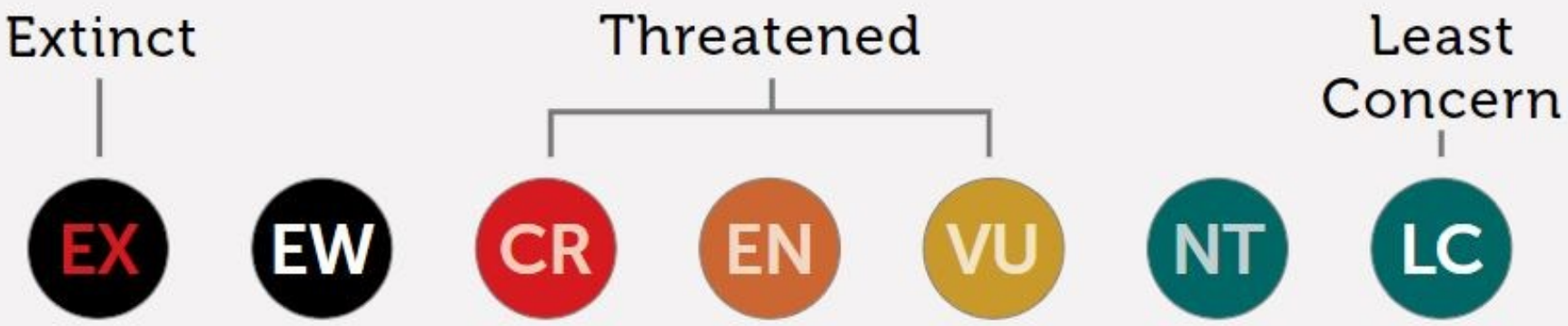
- **Blue Book:** The United Nations Environment Programme **(UNEP)** has compiled data on endangered species of the world in the Blue Book.

RED LIST CATEGORIES AND CRITERIA



Source: IUCN Red List

THE RED LIST CATEGORIES



Extinct (EX): no reasonable doubt that the last individual has died

Extinct in the Wild (EW): known only to survive in captivity, cultivation or well outside its natural range

Critically Endangered (CR): facing extremely high risk of extinction in the wild

Endangered (EN): facing a very high risk of extinction in the wild,

Vulnerable (VU): facing a high risk of extinction in the wild.

Near Threatened (NT): close to qualifying, or likely to qualify for a threatened category in the near future

Least Concern (LC): population is stable enough that it is unlikely to face extinction in the near future

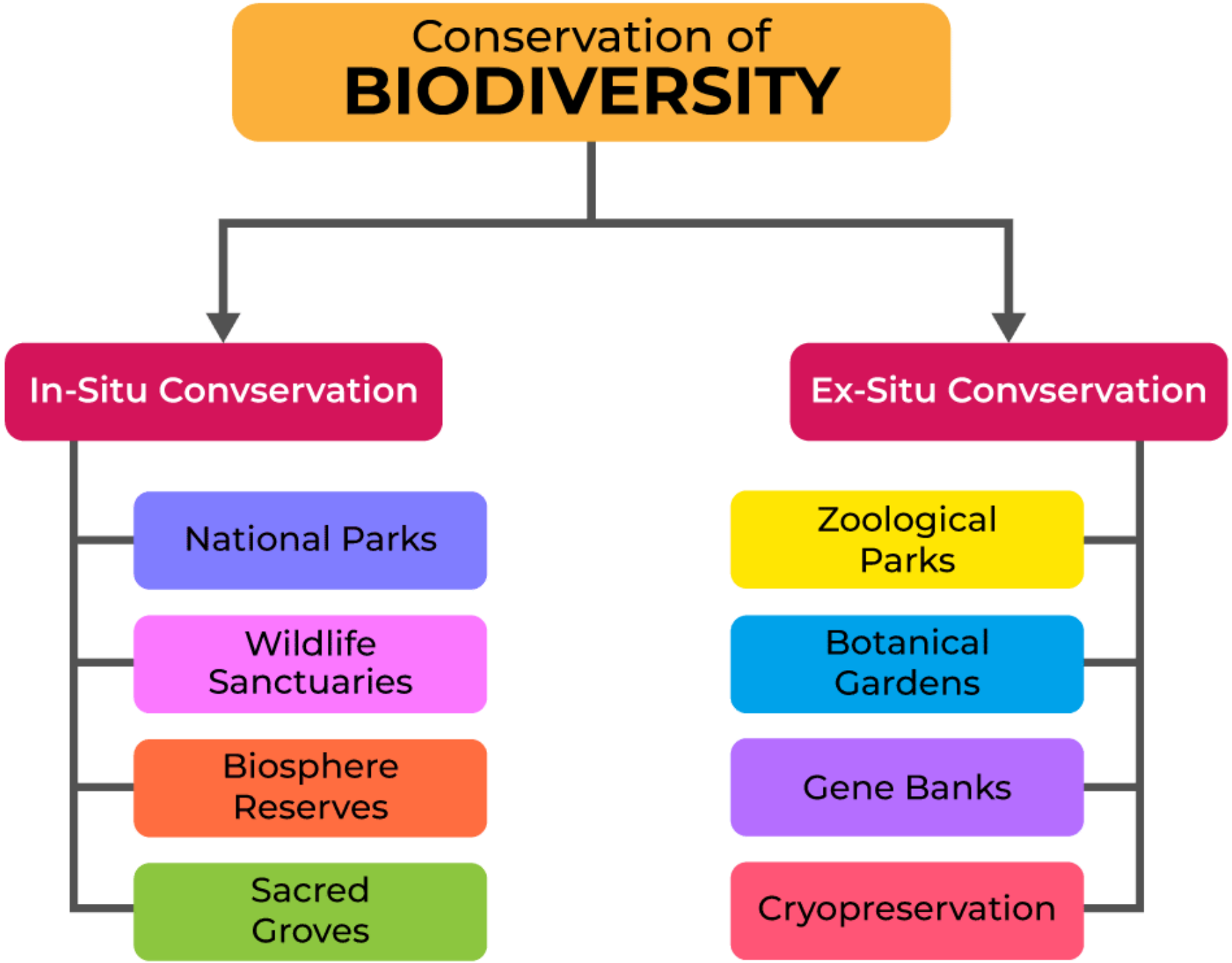
Data Deficient (DD): not enough information on abundance or distribution to estimate its risk of extinction

CONSERVATION OF BIODIVERSITY

Conservation is defined as management of human use of biosphere so that it may yield sustainable benefit to the present generation while maintaining its potential to meet the needs and aspirations of posterity.

There are two basic strategies of biodiversity conservation

- In-situ (on site)
- Ex-situ (off site)


Conservation of
BIODIVERSITY
In-Situ Convservation
National Parks
Wildlife Sanctuaries
Biosphere Reserves
Sacred Groves
Ex-Situ Convservation
Zoological Parks
Botanical Gardens
Gene Banks
Cryopreservation

A. IN-SITU CONSERVATION

- **Conservation of organism in its natural home through protection of a group of typical ecosystems by a network of protected areas, biosphere reserves, sacred forests and sacred lakes.**
- **Protected areas: such as National Parks and Wildlife Sanctuaries are areas of land and/or sea especially dedicated to the protection and maintenance of biological diversity, and of natural and associated cultural resources.**
- **There are more than 37000 protected areas around the world, while India has 612 protected areas (99 National Parks and 513 Wildlife Sanctuaries).**

NATIONAL PARKS

National park is a large areas managed for scientific educational and recreational use and are not usually used for commercial extraction of resources

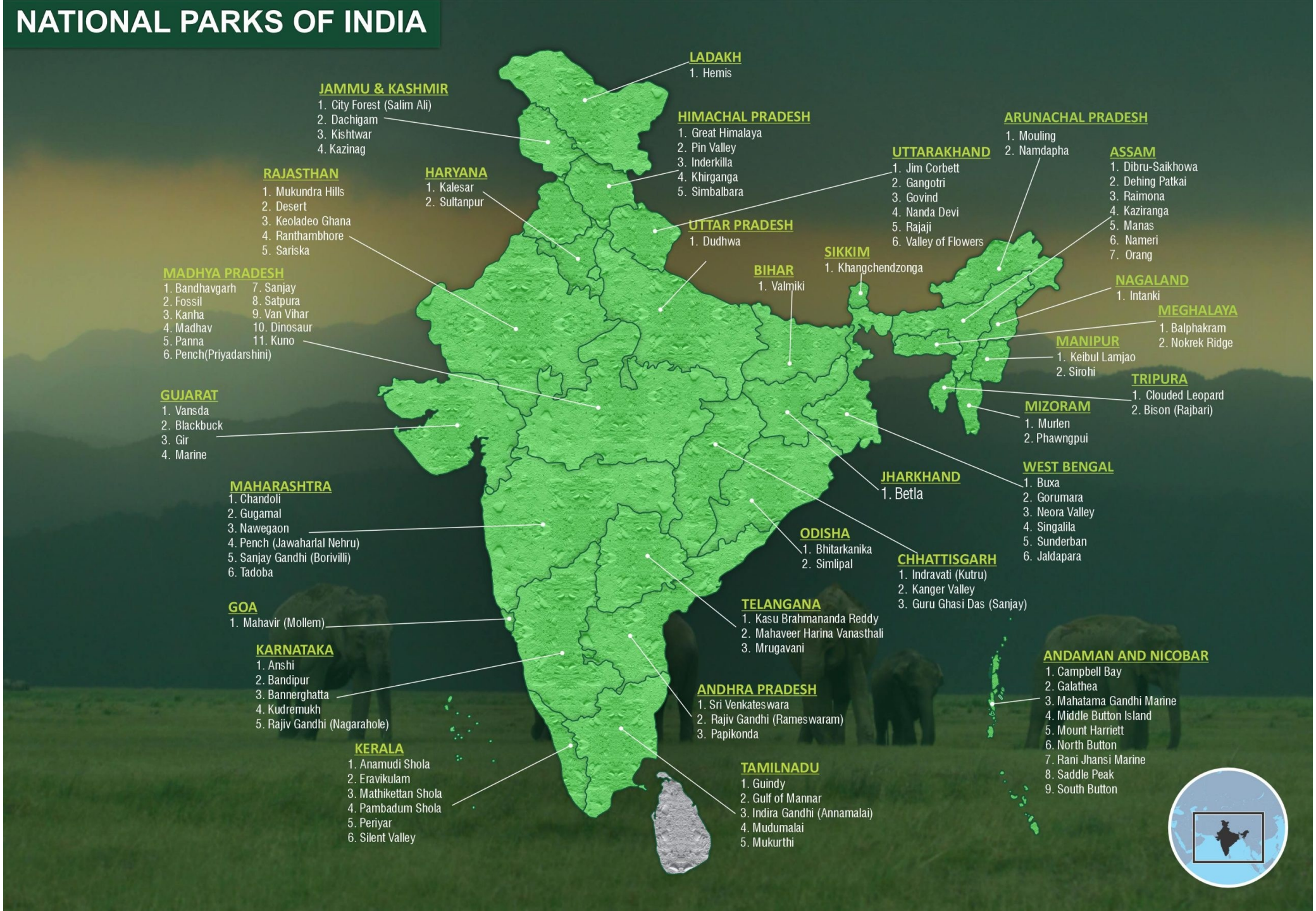
It is relatively large area where:

- *Ecosystems can not be altered by human activity.*
- *Govt. has taken step to prevent exploitation.*
- *Visitors are allowed under special conditions.*

India's first national park was established in 1936, now known as Jim Corbett National Park, in Uttarakhand.

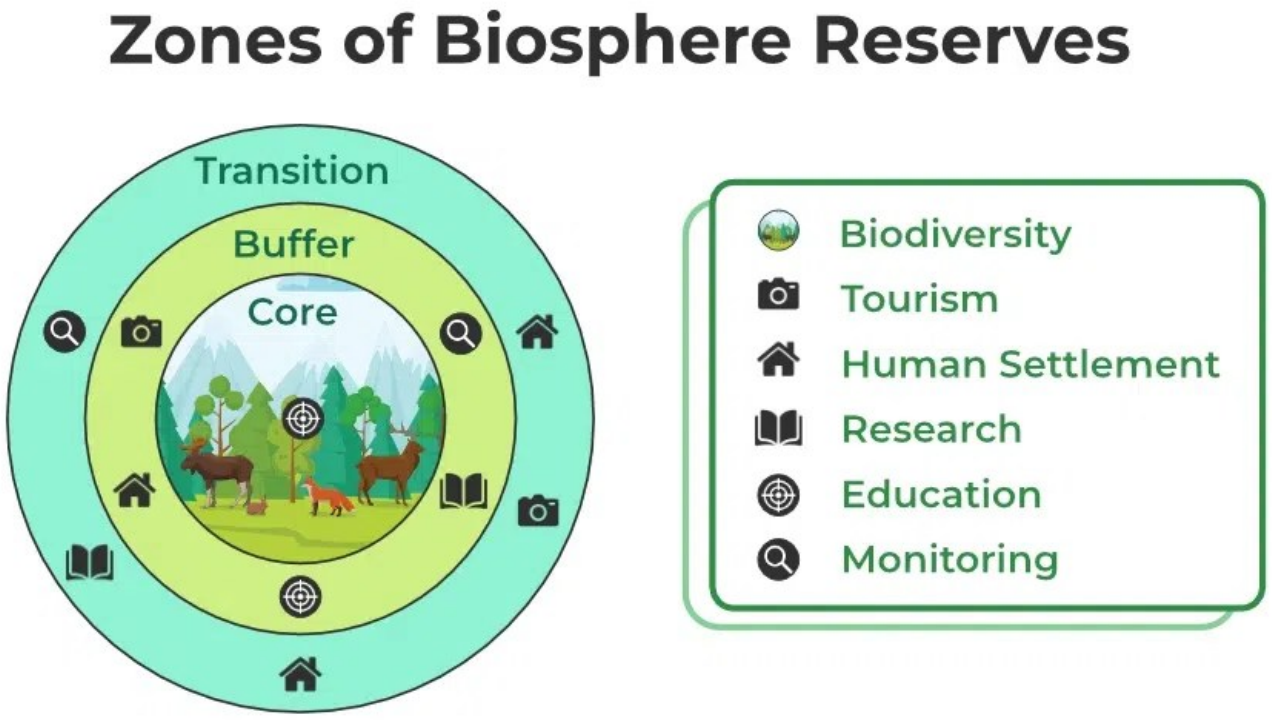
List of national parks of India

NATIONAL PARKS OF INDIA



BIOSPHERE RESERVE

- Biosphere reserves: are a special category of protected areas of land and/or coastal environments, wherein people are an integral component of the system.
- A biosphere reserve consists of core, buffer and transition zones.



- The core zone comprises an undisturbed and legally protected ecosystem.
- The buffer zone surrounds the core area and is managed to accommodate a greater variety of resource use strategies, research and educational activities.
- The transition zone, the outermost part of the biosphere reserve, is an area of active cooperation between reserve management and the local people. This is the place for activities like settlements, cropping, forestry and recreation and other economic uses continue in harmony with conservation goals.
- Objectives of Biosphere reserve:
  - *Conservation*
  - *Development*
  - *logistic*
- There are 738 biosphere reserves located in different countries out of which 18 biosphere reserves located in India.

WILDLIFE SANCTUARY

- A Wildlife Sanctuary is a protected area of importance for flora, fauna, or features of geological or other interest, which is reserved and managed for conservation and to provide opportunities for study or research. The Wild Life (Protection) Act, 1972 provides for the establishment of Protected Areas in India.
- It can be use for Study, Research and Tourism
- Caring Animals
- Educating the Public to treat animals

SACRED GROOVE

- Sacred groves of India are forest fragments of varying sizes, which are communally protected, and which usually have a significant religious connotation for the protecting community. Hunting and logging are usually strictly prohibited within these patches.
- Indian sacred groves are often associated with temples, monasteries, shrines, pilgrimage sites, or with burial grounds.
- Around 14,000 sacred groves have been reported from all over India, which act as reservoirs of rare fauna, and more often rare flora, amid rural and even urban settings.

B. EX-SITU CONSERVATION

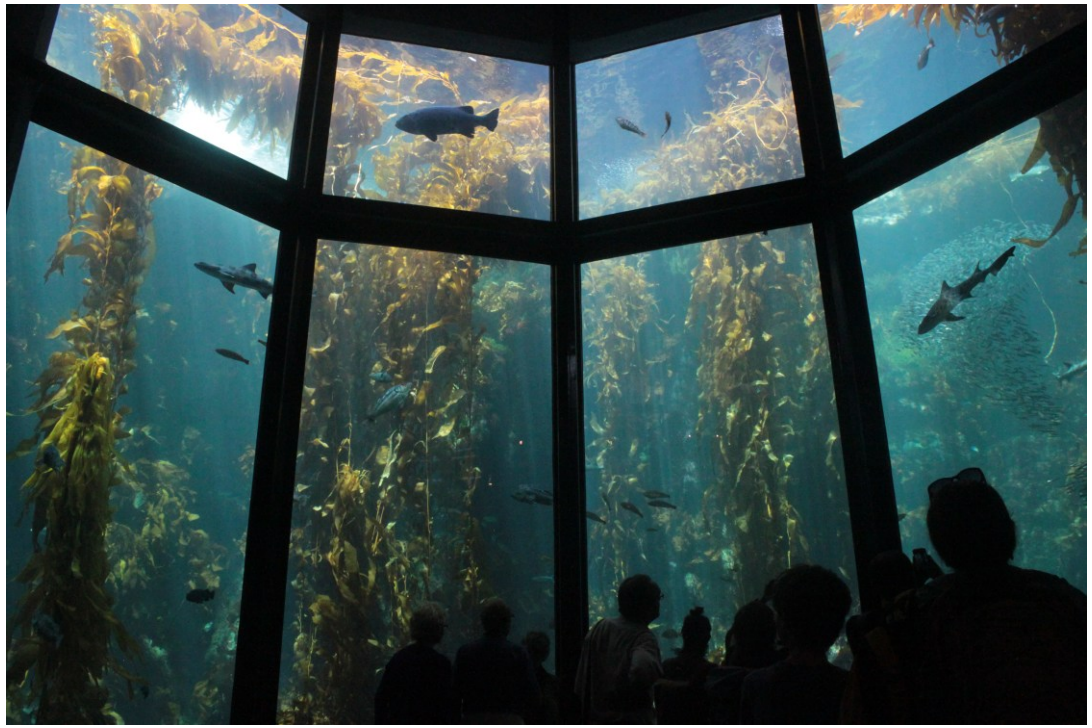
- Conservation of organisms in botanical gardens, zoos, conservation stands, gene, pollen, seed, seedling, tissue culture and DNA banks.
- There are more than 1500 botanical gardens in the world containing more than 80,000 species and many of these now have seed banks, tissue culture facilities and other ex situ technologies.
- Similarly there are more than 800 professionally managed zoos around the world with about 3000 species of mammals, birds, reptiles and amphibians.
- Many of these zoos have well developed captive breeding programmes.
- Plants and animals conserved in botanical gardens, arboreta, zoos and aquaria can be used to restore degraded land, reintroduce species into wild, and restock depleted populations.

ZOOS

Zoos around the world currently contribute to the conservation of nearly 7,000 species of terrestrial vertebrates (mammals, birds, reptiles, and amphibians) by caring for more than 500,000 individual animals. They do not do this alone; they often work with government agencies, universities, and a variety of other organizations who use zoo animals for research, education, and other conservation activities. While zoos traditionally focused on displaying charismatic animals that draw visitors, many zoos are now also investing in the conservation of small threatened vertebrates, as well as invertebrates, such as butterflies, beetles, dragonflies, spiders, and molluscs (many of which are also cheaper to maintain).

AQUARIA

Aquaria are the aquatic version of zoos, specialized in caring, displaying, and conserving marine and freshwater biodiversity, such as fishes, corals, molluscs, and crustaceans. Most organisms currently in aquaria have been obtained from the wild, but conservationists are constantly refining techniques to breed more species in captivity to limit wild collecting. Recent and dramatic increases in aquaculture, which currently accounts for roughly a third of fish and shellfish production globally, have made ex situ conservation of aquatic species even more important. The hope is that these ex situ populations will help maintain genetic stocks and act as insurances against disease outbreaks introduced by domestic fish, molluscs, and crustaceans.

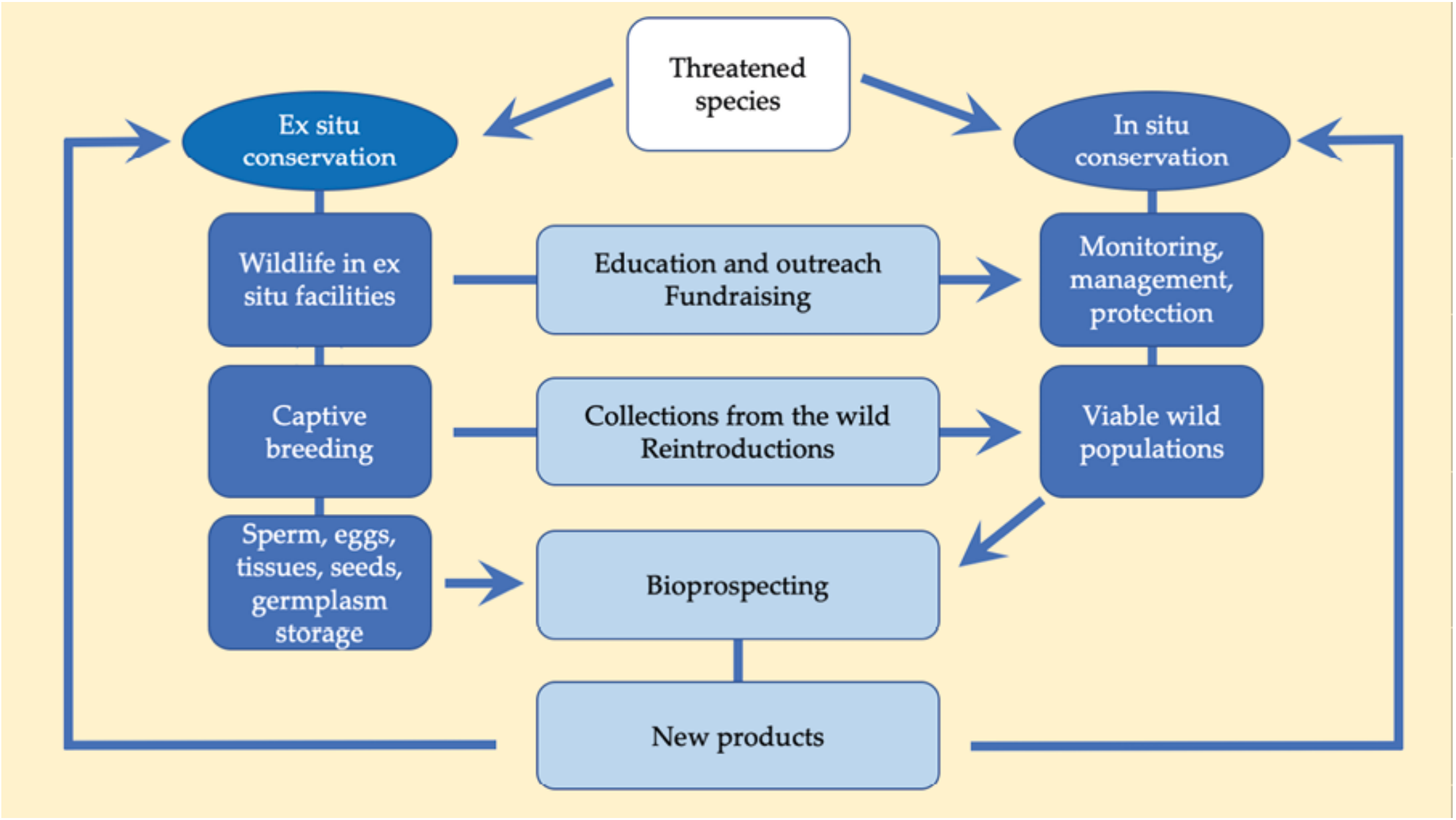


BOTANICAL GARDENS

- Botanical gardens (and arboretums, which specialize on trees and other woody plants) are dedicated to the collection, cultivation and educational curation of living plant species.
- Botanical gardens across the world house more than 6 million living plants, representing over 80,000 species—approximately 25% of the world's vascular flora (Wyse Jackson, 2001).
- The world's oldest and largest botanical garden—the **Royal Botanic Gardens in London**, UK—maintains over 28,000 plant taxa, nearly 10% of plant taxa in the world.
- Like zoos and aquaria, botanical gardens play a critical role in conservation efforts through public outreach and education.

GENE BANK

- Gene banks are a type of biorepository that preserves genetic material. For plants, this is done by in vitro storage, freezing cuttings from the plant, or stocking the seeds (e.g. in a seedbank). For animals, this is done by the freezing of sperm and eggs in zoological freezers until further need. With corals, fragments are taken and stored in water tanks under controlled conditions.
- Genetic material in a 'gene bank' is preserved in a variety of ways, such as freezing at -196 °C in liquid nitrogen, being placed in artificial ecosystems, or put in controlled nutrient media.



There are several ways in which in situ (on site) and ex situ (off-site) conservation can complement each other. No species conforms exactly to this idealised model, but nearly all species present some of these elements.

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